

Unit I

- 1) Cover all basic defn
- (1) max / min
 - (2) add / sub
 - (3) mult / div
 - (4) All test with eg.

2 Check conv of seq $\{a_n\}$

(i) $a_n = 1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n}$

(ii) $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$

3 Discuss conv of following:-

$$(i) \sum (-1)^{n-1} \frac{n}{5n+1} \quad \cdot \quad (ii) \sum (-1)^n \frac{n+5}{n(n+1)}$$

$$(iii) 2 - \frac{3}{2} + \frac{4}{3} - \frac{5}{4} + \dots$$

$$(4) \frac{1}{1 \cdot 2 \cdot 3} + \frac{3}{2 \cdot 3 \cdot 4} + \frac{5}{3 \cdot 4 \cdot 5} + \dots$$

$$(5) \sum \sqrt[3]{n^3+1} - n$$

$$(6) \sum \sqrt{n^2+1} - n$$

$$(7) \sum \frac{\sqrt{n}}{\sqrt{n^2+1}} x^n$$

$$(8) \frac{x}{1 \cdot 2} + \frac{x^2}{3 \cdot 4} + \frac{x^3}{5 \cdot 6} + \dots$$

$$(9) \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \dots$$

$$(10) \frac{1^2}{2^2} + \frac{1^2 \cdot 2^2}{2^2 \cdot 4^2} + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$(11) x + \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \dots$$

$$(12) 1 + \frac{3x}{7} + \frac{3 \cdot 6}{7 \cdot 10} x^2 + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13} x^3 + \dots$$

$$(13) \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} + \dots$$

4) Discuss conv of $\sum_{n=1}^{\infty} \frac{1}{n^p}$, $p > 1$

5) Discuss absolute conv of :-

(i) $\sum \frac{\sin n}{n^2}$ (ii) $\sum (-1)^{n+1} \frac{\cos n}{n \sin n}$

(iii) $\sum \frac{(-1)^{n+1}}{2n-1}$

(6) Discuss conv of (i) $\sum \frac{1}{n} \sin \frac{1}{n}$
(ii) $\sum \cot^{-1} n^2$

(7) $1 + \frac{2^1}{2!} + \frac{3^1}{3!} + \dots$

(8) $\sum \frac{4 \cdot 7 \cdot 10 \dots (3n+1)}{1 \cdot 2 \cdot 3 \dots n}$

(9) $\sum \frac{(n!)^2}{(2n)!} x^n$

(10) $\sum \frac{1}{n(\log n)^p}$

(11) Find Taylor series expansion of

(i) $\sin x$ about $x = \pi/2$

(ii) $\sin x$ about $x = \pi/4$

(iii) $\cos x$ about $x = \pi/2$

(iv) $\cos x$ about $x = \pi/4$

Unit 3

Beta-Gamma Ques

- 1) Define beta and gamma f^n .
- 2) Prove that beta f^n is symmetric.
- 3) Solve $\sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}}$
- ★ 4) Derive relationship b/w beta and gamma.
- ★ 5) Evaluate $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$ or $\sqrt{\frac{1}{2}}$
- ★ 6) Prove that $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left[\frac{p+1}{2}, \frac{q+1}{2}\right]$
- 7) Prove that $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \frac{\pi}{\sqrt{2}}$
- 8) $\int_0^{\infty} x^6 e^{-2x} dx$
- 9) Prove that $\int_0^{\infty} \frac{x^8 (1-x^6)}{(1+x)^{24}} dx = 0$
- 10) Find $\int_0^{\pi/2} \sin^3 x \cos^{\frac{5}{2}} x dx$
- 11) Simplify $\int_0^1 x^3 (1-x)^5 dx$
- 12) Simplify $\int_0^1 x^5 (1-x^3)^3 dx$
- 13) Find value of $\int_0^{\infty} e^{-x^2} dx$.

14) Solve i) $\int_0^3 x^3 (27 - x^3)^{-1/3} dx$ ii) $\int_0^1 x^2 (1 - x^3)^{2/3} dx$

15) Show that $\beta(m, n) = \beta(m, n+1) + \beta(m+1, n)$

16) Show that $\int_0^{\pi/2} \sin^7 \theta d\theta = \frac{16}{35}$

17) $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{1}{\sqrt{\sin \theta}} d\theta$. $\int_0^{\infty} \frac{y^2}{1+y^4} dy$

18

Unit-3 > Improper Integrals

- 1) Discuss types of improper integrals
- 2) Examine convergence of:-

★ (i) $\int_0^1 \frac{1}{\sqrt{x-x^2}} dx$

★ (ii) $\int_2^{\infty} \log x dx$

(iii) $\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}} dx$

★ (iv) $\int_{-\infty}^0 \frac{1}{a^2 + b^2 x^2} dx$

(v) $\int_{-\infty}^0 e^{4x} dx$

(vi) $\int_1^{\infty} \frac{\tan^{-1} x}{x^2} dx$

(vii) $\int_0^2 \frac{dx}{x^2 - 4x + 3}$

★ (viii) $\int_a^{\infty} \frac{1}{x^p} dx, p > 0$

- 1) Find length of arc of parabola $y^2 - 4y + 2x = 0$, lying in 1st quadrant.
- 2) Find length of curve $y = x^{2/3}$ from $x = -1$ to $x = 8$.

- 3) Find distance travelled from $t = 0$ to $t = \frac{\pi}{2}$ whose position is,

$$x = ca(\cos t + t \sin t)$$

$$y = ca(\sin t - t \cos t)$$

- 4) Find length of $x^{1/3} + y^{1/3} = a^{1/3}$

- 5) Find length of cycloid

$$x = a(\theta + \sin \theta) ; y = a(1 + \cos \theta)$$

- 6) Find volume of solid formed by revolving one arc of cycloid $x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$ about x -axis

- 7) Find volume of solid formed by revolving $r = a(1 + \cos \theta)$ about initial line.

- 8) Find area of surface formed by revolution of (i) $r = 2a \cos \theta$ about initial line
- (ii) $x^{1/3} + y^{1/3} = a^{1/3}$ about x -axis

(9) Solve length of curve $x=t^3, y=2t^2$
on $[0, 1]$

(10) Find surface area formed by
revolution of ellipse $x^2 + 4y^2 = 16$
about major axis.